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In-Situ PS/PMMA-2D Composites Microbeads as Potential Hydraulic Fracturing Proppants

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Abstract

In this report, we present copolymer composite microbeads (100/120 mesh) designed as in-situ high-strength proppants incorporating copolymer-2D nanofillers for effective hydraulic fracturing operations. The poly(styrene/methyl methacrylate)-2D nanofiller (PS/PMMA-2D nanofiller) composite microbeads were synthesized through an emulsion polymerization technique. Various 2D nanofillers, including commercial graphene (CG), hexagonal boron nitride nanosheets (h-BN), and a combination of CG and h-BN (CG:BN), were utilized in the preparation of these copolymer composite microbeads. The microbeads were thoroughly characterized using scanning electron microscopy (SEM) to analyze their morphology. Fourier-Transform Infrared (FT-IR) spectroscopy and X-ray diffraction (XRD) methods were also employed to confirm the development of the composites. FT-IR, XRD, and SEM findings indicated successful composite formation. Remarkably, the copolymer composite microbeads demonstrated crush resistance of up to 12,000 psi, suggesting their potential as effective candidates for hydraulic fracturing applications.

Introduction

Proppants play a critical role in maintaining the flow conductivity of fractures in the oil and gas industries. (Chen et al., 2022; Isah et al., 2021; F. Liang et al., 2016; Milton-Tayler and Duenckel, 2019; Sahai and Moghanloo, 2019). In general, proppants are spherical solids with high thermo-mechanical stability and salinity tolerance (Duenckel et al., 2017; Mader, 1989; Michael et al., 2020b). Though the proppants are primarily employed in the oil and gas industries, there are a few limiting factors that include the settling of proppants, which results in an early screen-out and less effective propped area, and leak-off of the fracturing fluid that reduces the width of the fracture (Anschutz et al., 2019; Liang et al., 2015; Tong and Mohanty, 2016; Yao et al., 2022). The aforementioned problems with conventional fracturing fluids and proppants are the primary reasons for the failure of the fracturing process. A high density of traditional proppants, such as quartz sand, causes the settling problem in fracturing fluid (Liu et al., 2015; Sun et al., 2022), resin-coated sands (Krishnan et al., 2022c, 2021a, 2021b; Li et al., 2023, 2022, 2021; Liu et al., 2023;

Zoveidavianpoor and Gharibi, 2015), metal oxide (Assem and Nasr-El-Din, 2017; Hellmann et al., 2014; Liu et al., 2016), and ceramics (Abd El-Kader et al., 2020; Anaya et al., 2022; Man and Wong, 2017; Wu et al., 2017). Therefore, it is highly preferable to develop light-weight proppants with lower settling velocities and higher transportation distance without compromising their thermal and mechanical stabilities and salinity tolerance (Song et al., 2021; Zoveidavianpoor et al., 2018). The density of the proppants certainly depends on their composition. By choosing the most suitable precursors and components, the density of the proppants can be easily controlled (Krishnan et al., 2022b; Tang et al., 2017). In this regard, there were many reports in the literature on polymer microspheres as successful light-weight proppants (Danso et al., 2021; Feng et al., 2021; Liao et al., 2022; Pangilinan et al., 2016; Tasqué et al., 2021; Yin et al., 2022). Polystyrene (PS), polymethyl methacrylate (PMMA), and their copolymer microbeads have mainly been studied for this purpose because of their chemical inertness, appreciable mechanical strength, and low cost (Chen et al., 2021; Krishnan et al., 2017; C. Liang et al., 2016; Song et al., 2021; Tasqué et al., 2021; Zhang et al., 2016; Zhou et al., 2017). Nevertheless, these polymer microbeads suffer from lower thermal stability. To improve the thermal stability and further enhance the mechanical stability of the polymer microbeads, adding inorganic fillers is a promising strategy (Krishnan et al., 2021a, 2021b). For instance, Chen et al. described heat-resistant PMMA proppants by incorporating silica fume microspheres (Chen et al., 2014), and graphite (Chen et al., 2015). Chen et al. also showed that incorporation of carbon black into PS microspheres would result in heat-resistant PS/carbon black composite proppants (Ma et al., 2019). Interestingly, the addition of nanofillers into the polymer matrix was shown to reinforce the mechanical strength of the polymers dramatically (Krishnan et al., 2022a, 2021c; Michael et al., 2020a, 2019). Chen et al. also showed that the crush resistance of PS/Graphite microbeads is improved mainly by the graphite nanofillers to 68 MPa (Han et al., 2014). It is well established that pre-formed solid proppant-laden fracturing fluids can cause significant abrasion to fracturing equipment, resulting in a markedly reduced service life. Moreover, the leakage of fracturing fluids and the settling of proppants can lead to early screen-outs and a less effective propped area. These issues associated with conventional pre-formed proppants are primary contributors to fracturing failures. In-situ generated proppants (IGPs), which utilize the reservoir's high-temperature and high-pressure (HT-HP) conditions for a liquid-to-solid phase transition, are highly sought after as a solution to the drawbacks of pre-made proppants (Krishnan et al., 2024; Li et al., 2024).

In-situ proppants, formed within the fracture during hydraulic fracturing, are primarily used in unconventional reservoirs, such as shale formations, where the low permeability and tight nature of the rock require fracture stimulation (Xu et al., 2025). While their use is also applicable in conventional reservoirs, its significance is heightened in unconventional settings to enhance fracture conductivity and increase hydrocarbon production. In conventional reservoirs, proppants are commonly employed to increase the production from naturally fractured formations or to stimulate output in zones with lower permeability. However, the fractures in conventional reservoirs are generally larger and more conductive than those found in unconventional reservoirs, and therefore, the use of proppants in conventional fractures is not critical (Rylance et al., 2025; Shafiq et al., 2025). Unconventional reservoirs, including shale and tight sand formations, exhibit extremely low permeability and porosity. Hydraulic fracturing is crucial for establishing conductive pathways that facilitate the flow of oil and gas from the reservoir to the wellbore. In-situ proppants, generated through various chemical or physical processes within the fracture, are essential for maintaining these fractures and optimizing the flow of hydrocarbons. In the continued effort to further improve the thermo-mechanical strength of lightweight polymer proppants, the current study reports the preparation of PS/PMMA copolymer composite microbeads with different 2D nanofillers, including commercial graphene (CG), hexagonal boron nitride nanosheets (hBN), and a mixture of CG and hBN. The PS/PMMA copolymer-2D nanofiller composite microbeads were prepared by the *in situ* emulsion polymerization method. The PS/PMMA composite microbeads exhibited very high crush resistance and

thermal stability, which have the potential to be successfully applied in hydraulic fracturing operations. The in-situ generation of copolymer composite proppants presents a significant advancement in the field of hydraulic fracturing. These proppants are formed directly within the fractures of the geological formation under specific down-hole conditions, thereby enhancing their adherence and efficiency. However, this in-situ polymerization process is contingent upon reaching a critical temperature threshold; specifically, it requires temperatures of 70 °C (158 °F) or higher for the polymer chains to form and interlink effectively. In cold reservoirs, where temperatures are insufficient to initiate this chemical reaction, the successful formation of proppants may not be achievable. This limitation underscores the importance of temperature management in the development and application of these proppant materials, particularly in enhancing production rates and the overall efficiency of resource extraction.

Experimental

To prepare the PS/PMMA-2D nanofiller composite microbeads, 100 ml of de-ionized water (DI) is taken in a 3-neck RB flask, which is connected to a condenser at one neck and an N₂ purging line at the other neck. The DI water is thoroughly purged with N₂ gas for 10 minutes to remove the dissolved oxygen in the water. Then, 500 mg of SDS was added and stirred for 30 minutes for complete dissolution. To which a pre-made mixture of 10 ml of 1:1 mixture of co-monomers (5 ml each of S and MMA), BPO (100 mg), and 2D-nanofiller (100 mg of CG or hBN or CG:BN) is added drop-wise with continuous stirring (450 rpm) and N₂ flow (1ml/min.). The temperature is increased to 70 °C to initiate the polymerization process. The formed PS/PMMA-2D nanofiller composite microbeads were collected after 8 hours of reaction (Fig. 1). The microbeads were thoroughly washed with DI water and ethanol multiple times to remove the surfactant (SDS) and any residual reaction components, and finally dried in a vacuum oven at 40 °C for 12 hours. The PS/PMMA microbeads were prepared similarly without adding the 2D nanofiller.

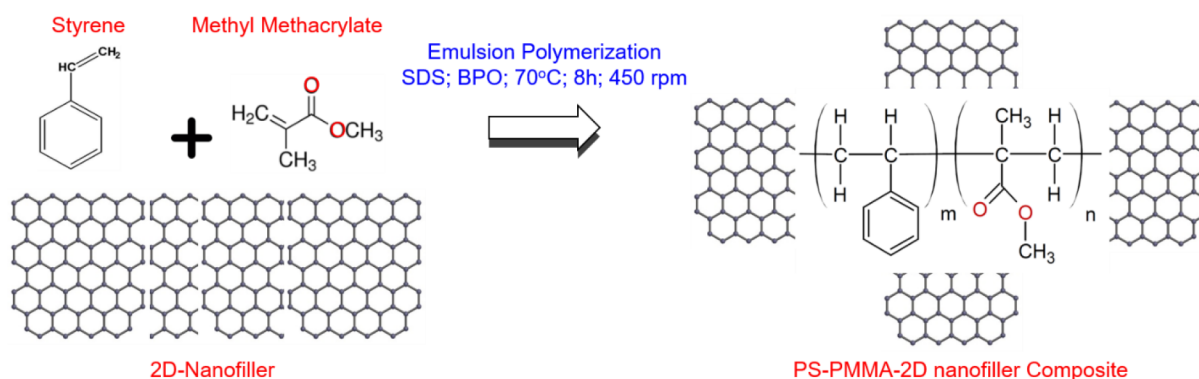


Figure 1—Schematic illustration of the preparation of PS/PMMA-2D composite microbead proppants

Results and Discussion

Fig. 2(b-d) show SEM images of the PS/PMMA-2D nanofiller composite (with CG, hBN, and CG:hBN, respectively) microbeads in comparison to PS/PMMA microbeads (Fig. 2a). It is evident from Figure 2 that the composite formation is successful, and the prepared samples seem to be perfectly spherical with sphericity and roundness values higher than 0.9. These results are also consistent with the optical images of the samples, as shown in. The higher sphericity and roundness values are crucial for any proppant, as they play a significant role in increasing the flow conductivity of the fractures. Note that quartz or resin-coated sand has a sphericity and roundness value higher than 0.6. Therefore, the PS/PMMA-2D nanofiller composite microbeads can exhibit higher flow conductivity than conventional sand or modified sand-based proppants.

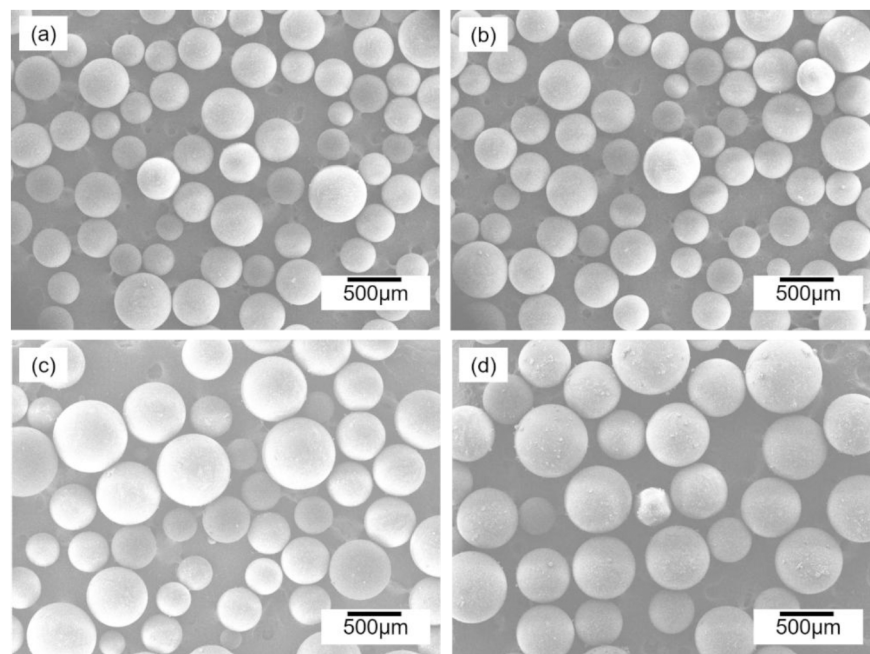


Figure 2—SEM images of (a). PS/PMMA, (b). PS/PMMA-CG, (c). PS/PMMA-hBN, and (d). PS/PMMA-CG:hBN

The chemical composition and the composite formation of PS/PMMA and PS/PMMA-2D nanofillers composite microbeads were investigated using FT-IR spectra. Fig. 3a shows the characteristic peaks for PS and PMMA. The peaks are detected at 691 cm^{-1} (the mono-substituted ring in-phase bending in PS), 750 cm^{-1} (asymmetric band of C-C-O group in PMMA), at 1150 cm^{-1} (stretching mode of C-O-C), at 1396 cm^{-1} and 2923 cm^{-1} (bending and stretching modes of the $-\text{CH}_3$ group, respectively), at 1616 cm^{-1} (stretching mode of C=C bond in the PS) and 1758 cm^{-1} (the C=O in the PMMA), which corresponds to the PS/PMMA copolymer microbeads. In addition, the characteristic peaks of CG (1074 cm^{-1} , and 1380 cm^{-1} (C=C stretching), 1627 cm^{-1} (C-O-C stretching at the surface of CG), 1722 cm^{-1} (C-C stretching)) and hBN (780 cm^{-1} (B-N bending), 1160 cm^{-1} (B-N) and 1336 cm^{-1} (B-N stretching)) peaks were also detected in the respective composites. Based on the FT-IR results, the formation of PS/PMMA-2D nanofillers composite microbeads was confirmed.

Fig. 3b shows the XRD of PS/PMMA-2D nanofiller composite microbeads compared to PS/PMMA microbeads. The PS/PMMA exhibited a broad peak with very low intensity, corresponding to its non-crystalline nature. For the PS/PMMA-CG composite microbeads, a broad peak with relatively higher intensity, accompanied by a sharp spike at 26° , is observed. The sharp peak at 26° is attributed to the (011) plane of graphene nanosheets. Similarly, in the PS/PMMA-hBN composite microbeads, a broad peak with relatively higher intensity, accompanied by a sharp spike at 26.5° , is noted. This peak corresponds to 110 planes of hBN. Interestingly, the PS/PMMA-CG:hBN composite microbeads exhibited a broad peak corresponding to the characteristic peaks of PS/PMMA, CG, and hBN. These results indicate the successful incorporation of the 2D nanofillers into the PS/PMMA copolymer matrix.

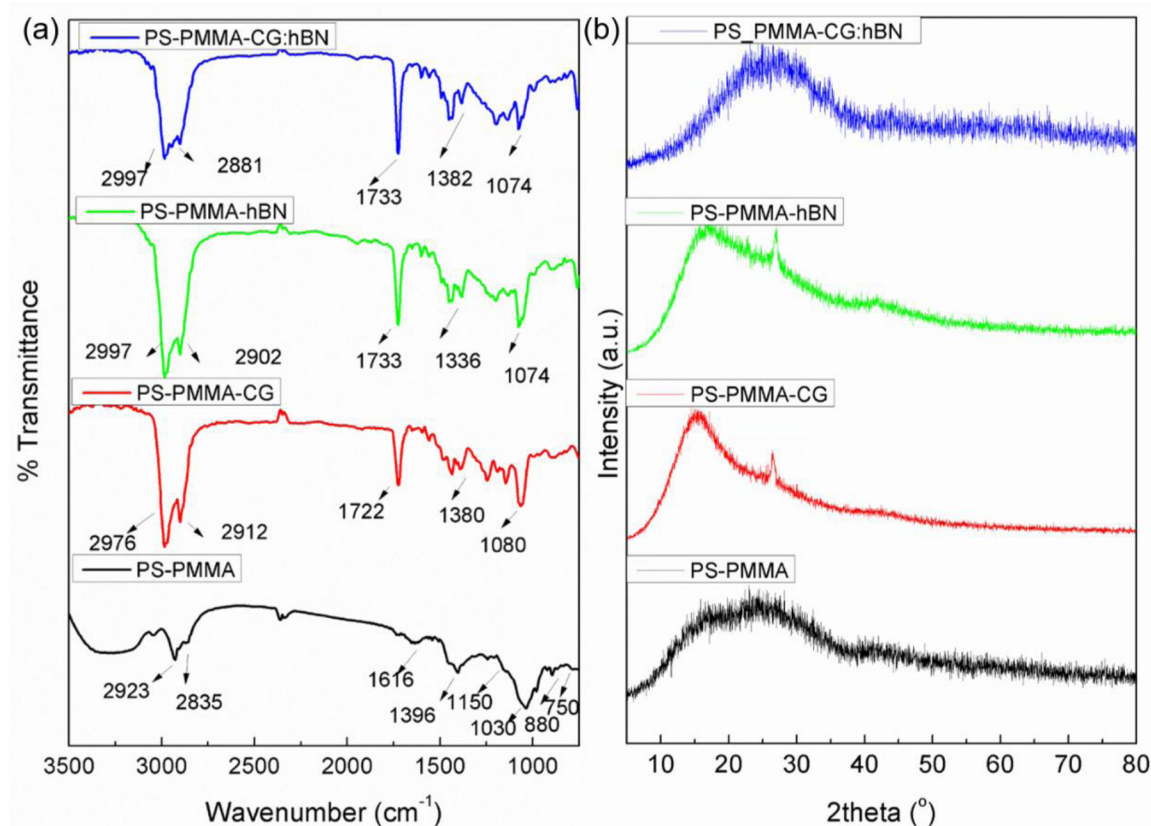


Figure 3—(a). FT-IR and (b). XRD of PS/PMMA-2D composite microbead proppants

Fig.4 presents the comprehensive results of the crush test conducted on various polymer microbead proppants, specifically PS/PMMA-CG, PS/PMMA-hBN, and PS/PMMA-CG:BN. These results are contrasted with those obtained from PS/PMMA microbeads under a range of applied stresses from 6000 psi to 14000 psi. As the applied stress increases, there is a corresponding rise in the percentage of fine particle generation across all types of polymer microbead proppants. Within the applied stress range of 6000 psi to 12000 psi, both PS/PMMA and the composite microbeads infused with 2D nanofillers exhibited remarkably low levels of fine production. Specifically, the tests revealed a fine generation of 1% at 6,000 psi, 2.5% at 8,000 psi, below 5% at 10,000 psi, and less than 10% at 12,000 psi. The findings illustrate that the fine particle production is nearly equivalent across these proppants, suggesting that the PS/PMMA and PS/PMMA-2D nanofiller composite microbeads maintain their integrity and performance even under the elevated pressures typically encountered in downhole conditions, up to 12000 psi. To delve deeper into the impact of the 2D nanofillers on the mechanical robustness of the PS/PMMA-2D nanofiller composite microbeads, the applied stress was escalated to 14000 psi. At this heightened level of stress, a notable distinction in mechanical stability was revealed between the composite microbeads and the standard PS/PMMA microbeads. Specifically, at 14000 psi, the PS/PMMA microbeads demonstrated a fine production of 20%. In contrast, the PS/PMMA-CG microbeads produced only 15% fine particles. Under identical test conditions, the PS/PMMA-hBN variant yielded a fine particle output of 14%, while the PS/PMMA-CG:hBN combination exhibited an even lower fine particle production of 12%. These outcomes highlight the enhanced mechanical stability achieved through the incorporation of 2D nanofillers in the composite microbeads, indicating their potential for superior performance in high-stress environments.

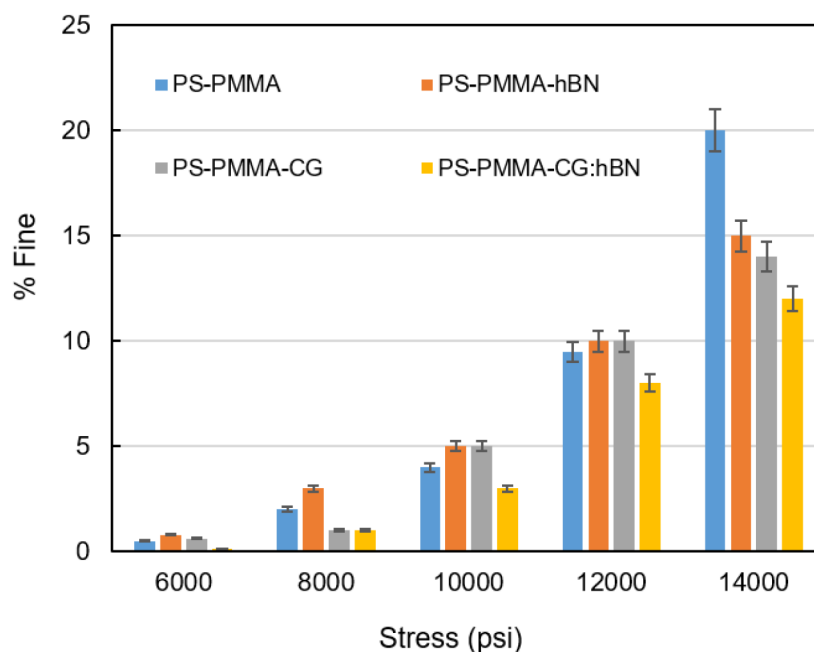


Figure 4—Crush test results of PS/PMMA-2D composite microbead proppants

Conclusion

In this study, we successfully synthesized copolymer composite microbeads featuring a precise mesh size of 100/120, demonstrating their promising application as high-strength proppants for hydraulic fracturing operations. The microbeads, composed of polystyrene (PS) and poly(methyl methacrylate) (PMMA), were created through an emulsion polymerization method, integrating various nanofillers, namely carbon graphite (CG), hexagonal boron nitride (h-BN), and a combination of both, to enhance their mechanical properties. Scanning Electron Microscopy (SEM) analyses revealed the distinct spherical morphology of the copolymer composites, confirming the successful formation of uniform particles. Further characterization through Fourier Transform Infrared Spectroscopy (FT-IR) provided insights into the chemical bonding and functional groups present in the microbeads, thereby affirming their complex chemical composition. X-ray diffraction (XRD) studies indicated a significant degree of crystallinity within the composite microbeads, which is crucial for their strength and stability under stress. Remarkably, the PS/PMMA-CG:hBN composite microbeads exhibited exceptional crush resistance, withstanding pressures of up to 12,000 psi. This substantial level of crush resistance underscores the microbeads' capability to endure the extreme conditions of hydraulic fracturing, thereby highlighting their potential as effective proppants in enhancing oil and gas extraction processes. The findings from this research not only contribute to the development of advanced materials for fracturing applications but also pave the way for further innovations in the field of material science and engineering.

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